

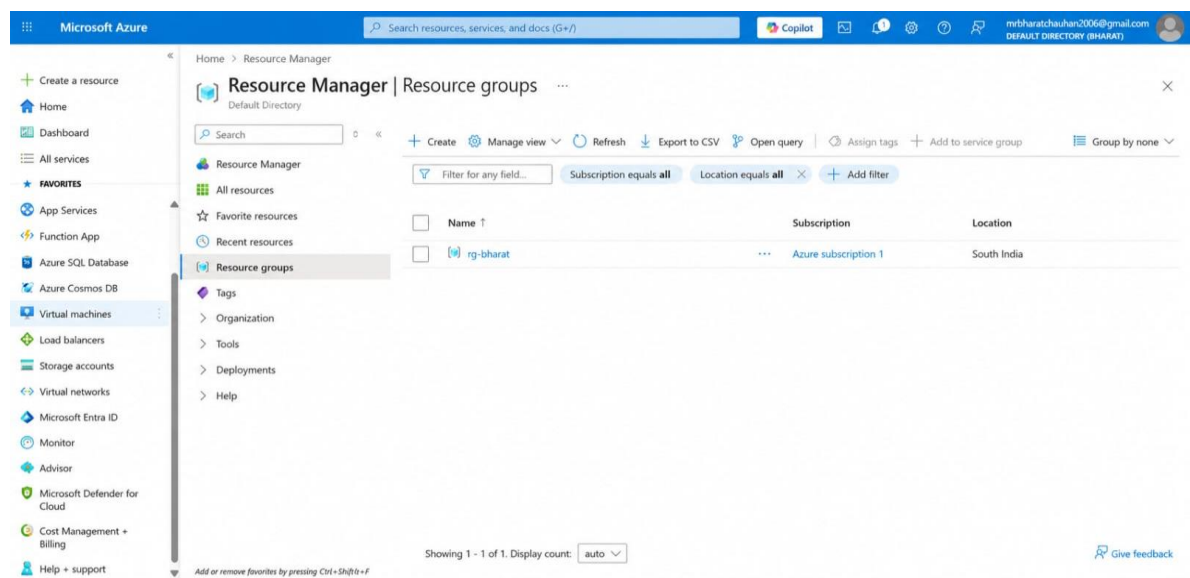
WEEK 4 ASSIGNMENT

INTRODUCTION

This assignment demonstrates the implementation of Azure Blob Storage and Azure Data Factory for copying data from one storage container to another. It also includes IAM role assignment and the use of the Get Metadata activity for retrieving file information before data movement.

TASK 1 – Create Resource Group

- Resource Group Name: rg-bharat
- Region: South India



Task 2 – Create Storage Account

Configuration

- Storage Account Name: bharatstorage12345
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)
- Region: South India

Home > bharatstorage12345_1781438754328 > Overview > bharatstorage12345

bharatstorage12345 | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix Only show active containers

Showing all 2 items

	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	6/14/2026, 5:39:36 PM	Private	Available	-
☐	input	6/14/2026, 5:41:26 PM	Private	Available	-

Upload Source CSV File

Home > bharatstorage12345_1781438754328 > Overview > bharatstorage12345 > Containers > input

input

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease

Authentication method: Access key (Switch to Microsoft Entra user account)

Add Filter

Search blobs by prefix (case-sensitive) Only show active blobs

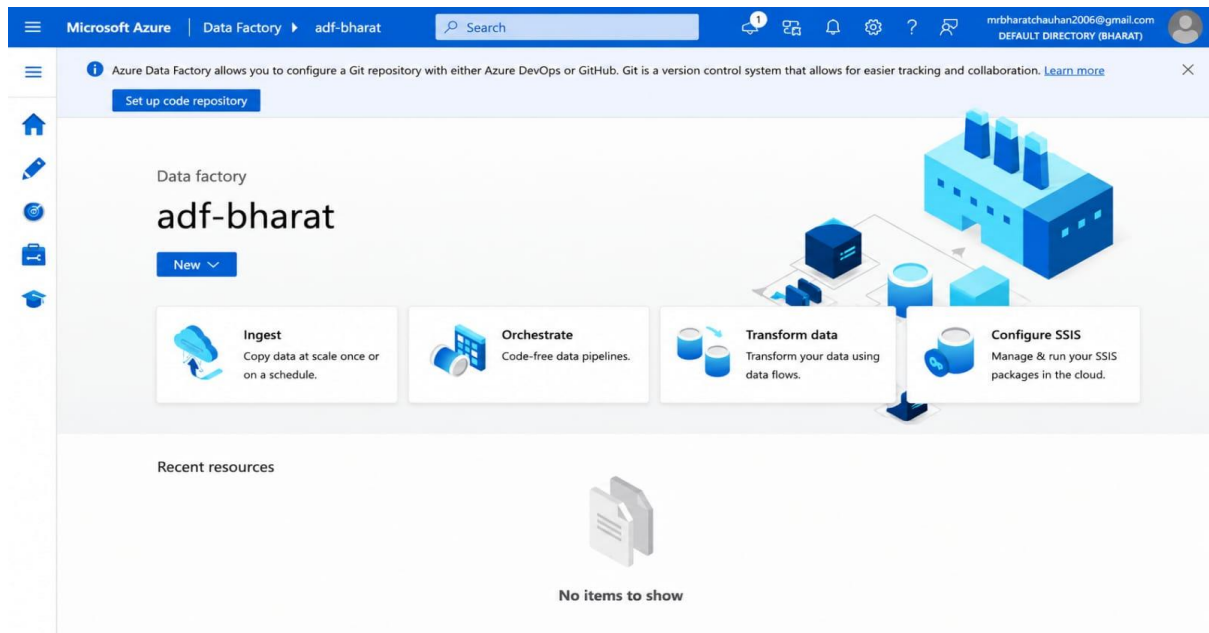
Showing all 1 items

	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample_Superstore...	6/14/2026, 5:44:39 PM	Hot (Inferred)	Block blob	21.39 KB	Available

TASK 3 – ADF Basics

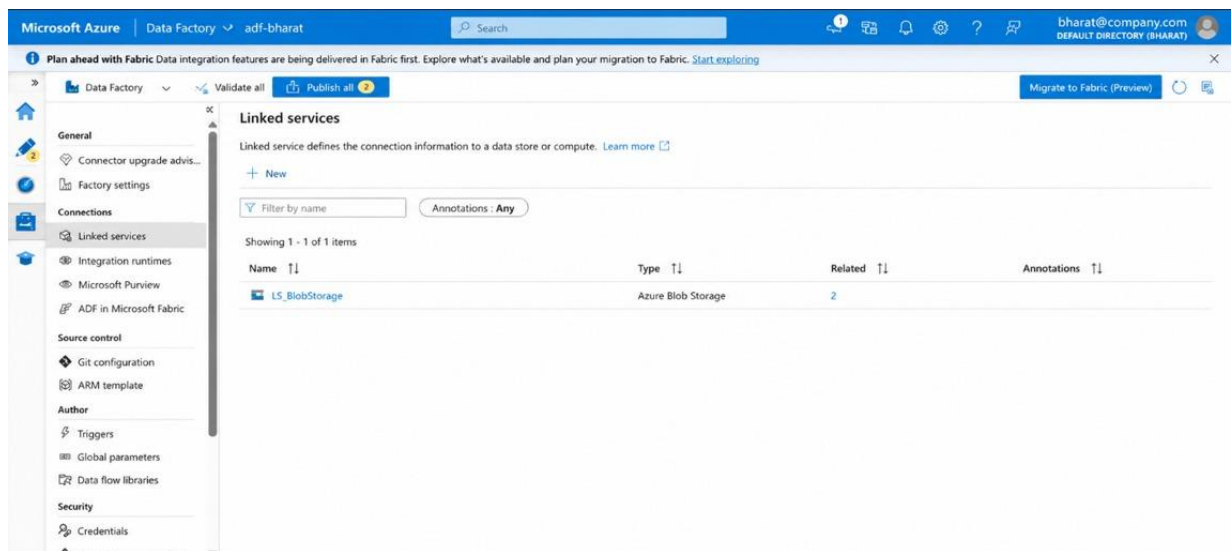
☐ Creation of Azure Data Factory

Azure Data Factory is a cloud-based data integration service provided by Microsoft Azure. It enables users to create, schedule, and manage data pipelines that move and transform data across different storage systems and services.



☐ Create Linked Service

A Linked Service acts as a connection between Azure Data Factory and external data sources. In this assignment, a Linked Service named LS_BlobStorage was created to establish a connection between Azure Data Factory and the Azure Blob Storage account. This connection allows the pipeline to securely access files stored in the storage account.



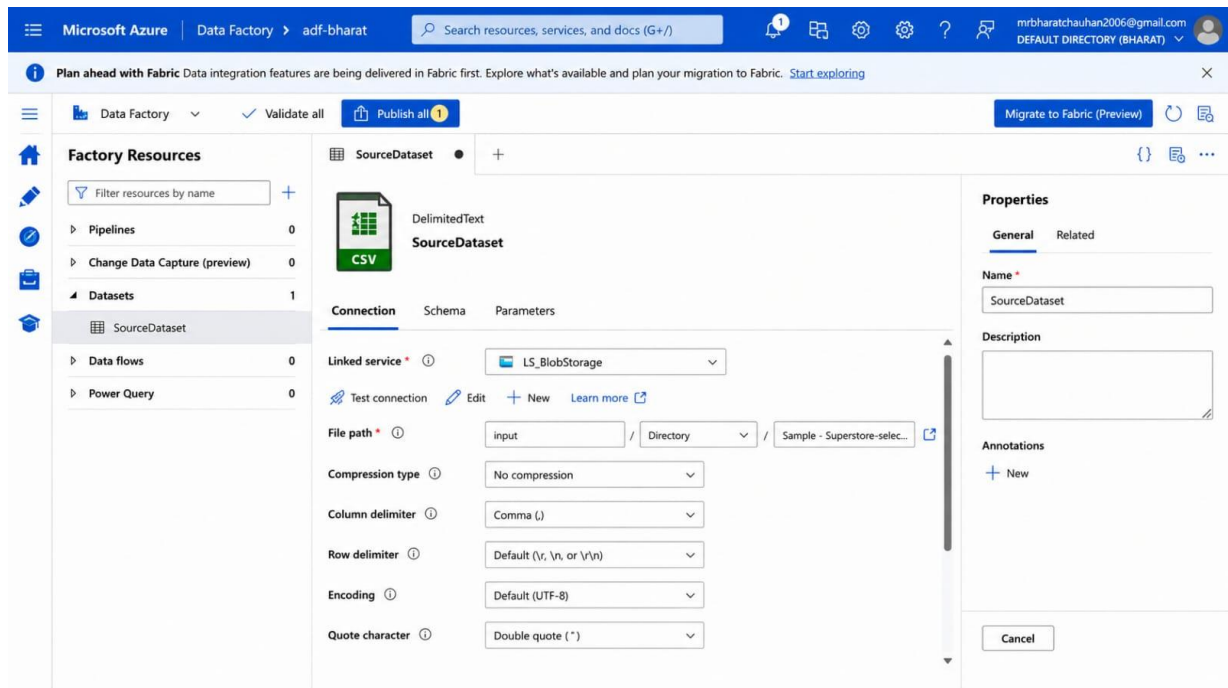
☐ Create Source and Destination Datasets

A Dataset represents the structure and location of data that will be used in a pipeline.

- Name: SourceDataset

- Container: input
- File: Sample - Superstore.csv

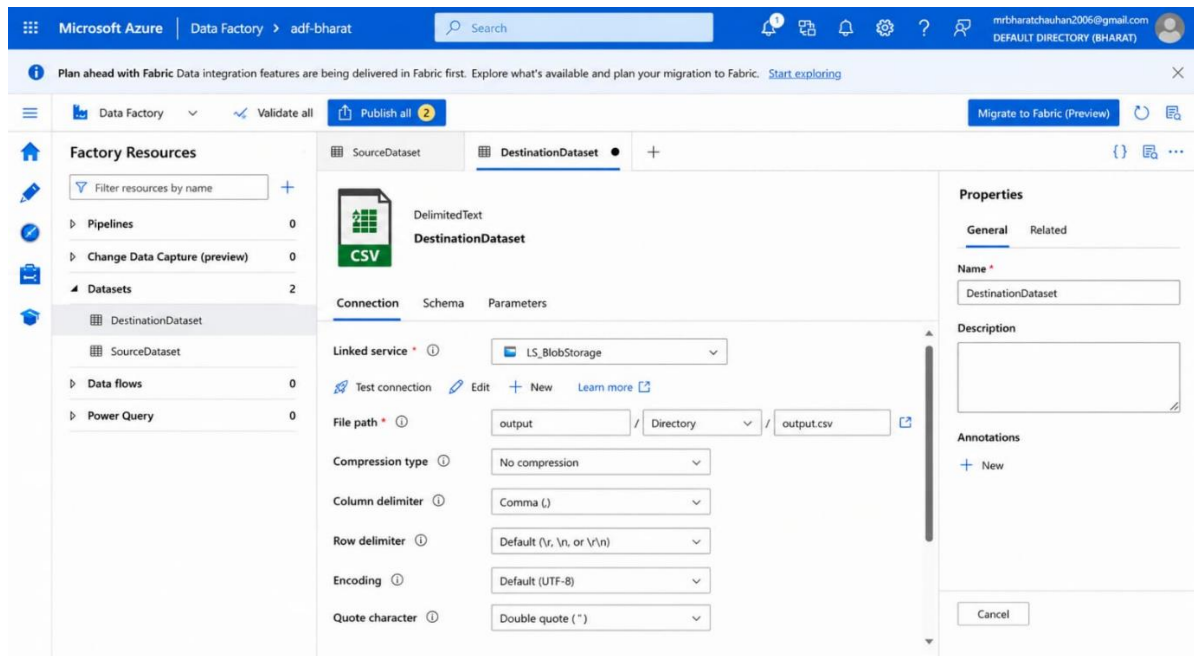
This data represent the input file that will be read by the pipeline.



Destination Dataset

- Name: DestinationDataset
- Container: output
- ☐ File: output.csv

This dataset represents the destination where the copied data will be stored.



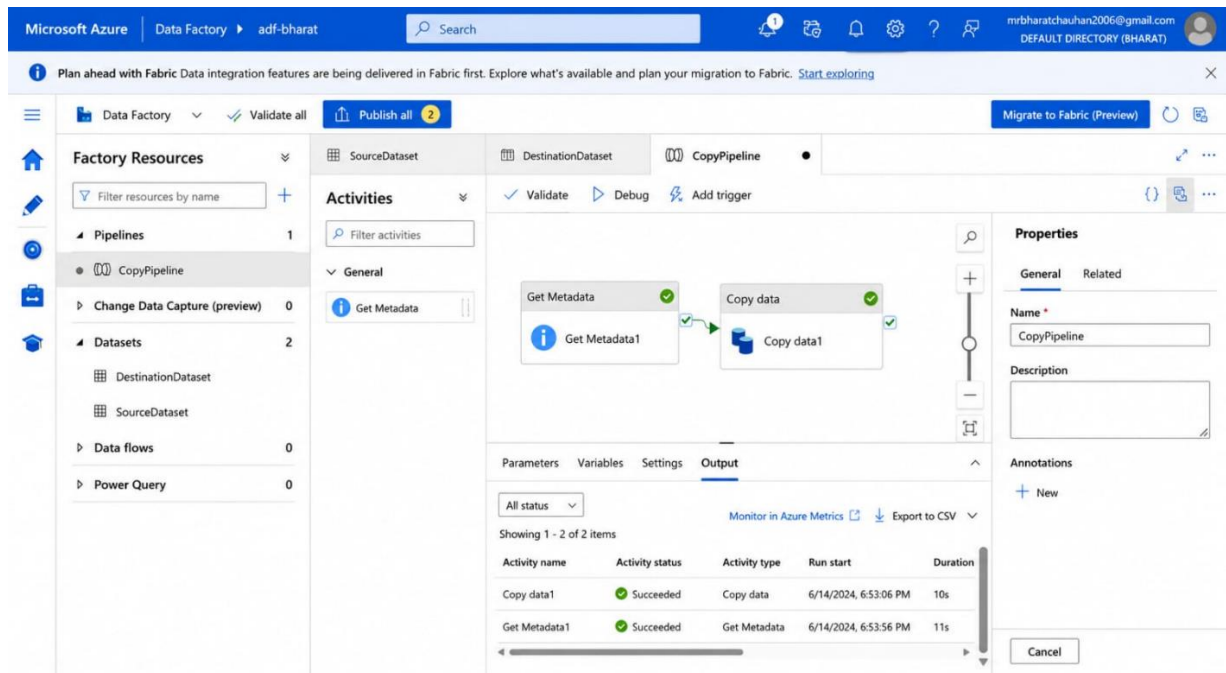
Use Get Metadata Activity

The Get Metadata activity is used to retrieve metadata information about a file or dataset before performing further operations.

In this assignment, the activity was configured with the SourceDataset and the Exists property was selected to verify whether the source file is available before executing the copy operation.

Purpose of Get Metadata

- Check whether the source file exists.
- Validate the availability of data before processing.
- Improve pipeline reliability by avoiding execution errors.



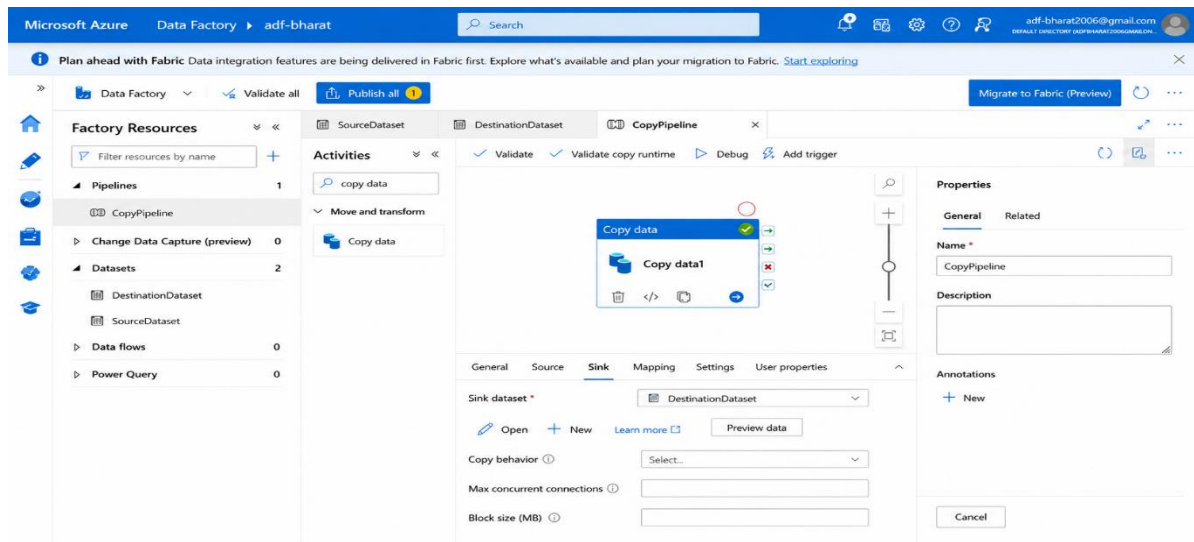
Task 4: Pipeline Development

Objective

The objective of this task is to create an Azure Data Factory pipeline that copies data from a source dataset to a destination dataset using the Copy Data activity.

Pipeline Development

A pipeline named CopyPipeline was created in Azure Data Factory. The Copy Data activity was configured to read data from the SourceDataset (stored in the input container) and copy it to the DestinationDataset (stored in the output container).



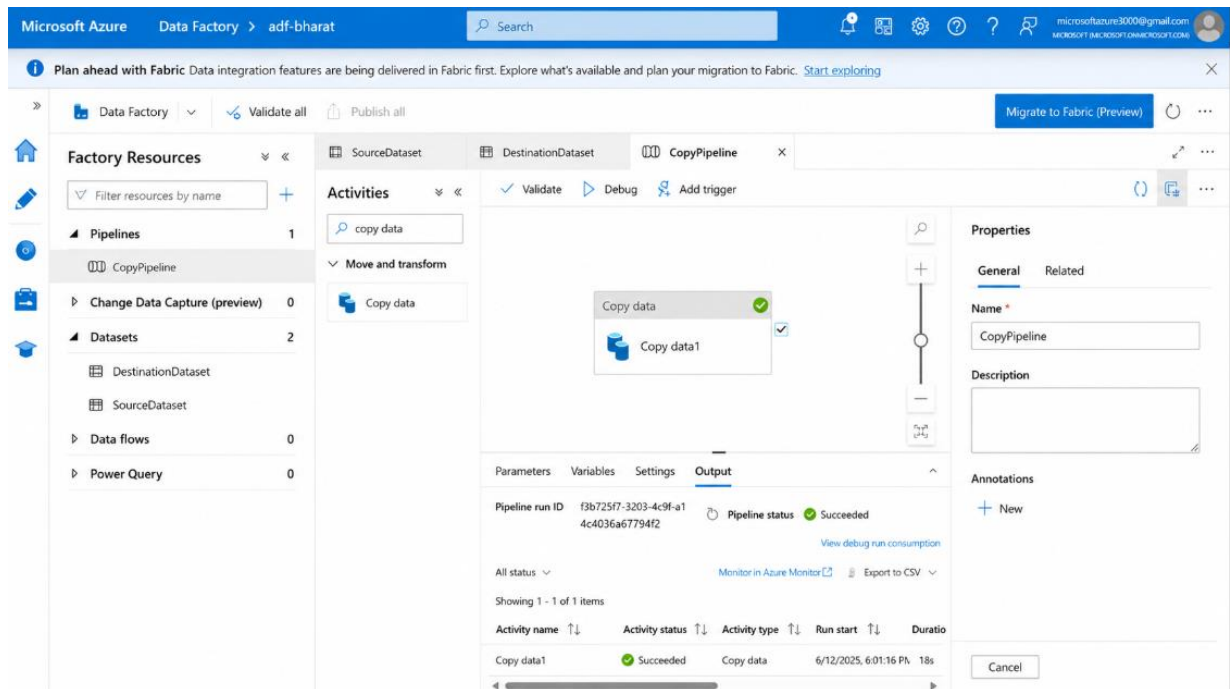
Task 5: Pipeline Execution

Objective

The objective of this task is to execute the Azure Data Factory pipeline and verify that the data transfer process completes successfully.

Pipeline Execution

The pipeline was executed using the Debug option in Azure Data Factory. During execution, the pipeline copied the data from the SourceDataset in the input container to the DestinationDataset in the output container.



Task 6: IAM Roles

Objective

The objective of this task is to manage access permissions by assigning appropriate IAM (Identity and Access Management) roles to users and Azure Data Factory.

Role Assignment

The Reader role was assigned to provide read-only access to the storage resources. Additionally, the Storage Blob Data Contributor role was assigned to Azure Data Factory to allow it to read and write data in Azure Blob Storage, enabling successful pipeline execution.

